SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2– Agile Sprint Planning & User Story Workshop

Course Code:	CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Code of Conduct

I Jai Ganesh Rao N/ 192524073 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



1. PROJECT VISION AND STAKEHOLDERS

1.1 Project Vision

The vision of the project is to develop an AI-assisted Dark Web Threat Actor Intelligence & Attribution Platform that helps authorized cybersecurity investigators collect, organize, and analyze information about threat actors operating on the dark web. The system should identify possible relationships between different online personas and present them with a clear attribution confidence level.

The system will reduce manual investigation effort by maintaining handles, PGP keys, wallet addresses, and infrastructure indicators in a centralized and secure database. Investigators can view threat-actor profiles, search across identifiers, follow activity on a timeline, and explore relationships in a graph. Overall, the system aims to improve the speed, accuracy, and reliability of threat-actor investigations while ensuring that similarities between personas are never treated as proof of identity.

1.2 Project Objectives

The main objectives of the project are:

- To collect threat-actor intelligence from multiple authorized dark-web sources.
- To maintain handles, usernames, PGP keys, wallet addresses, and other identifiers.
- To create and manage threat-actor profiles.
- To identify relationships between different online personas.
- To detect infrastructure-related indicators associated with hidden services.
- To apply AI-based stylometric and behavioural analysis to identify possible persona links.
- To assign an attribution confidence level to identified relationships.
- To provide a searchable relationship graph and an investigation timeline.
- To support autonomous and continuous intelligence collection.
- To allow investigators to export results in CSV, JSON, and report formats.
- To protect sensitive intelligence through access control, encryption, and audit logging.
- To provide useful reports for investigators and security teams.

1.3 Key Stakeholders

The major stakeholders involved in the project are:

Stakeholder	Role
Cybersecurity Investigators	Collect intelligence, analyze threat actors, and review relationships and confidence scores
Law Enforcement Agencies	Conduct lawful investigations and require reliable, well-documented evidence

Stakeholder	Role
Threat Intelligence Analysts	Collect, correlate, and verify intelligence from different sources
Security Researchers	Study threat-actor behaviour, personas, and identity changes
Organizations / Incident Response Teams	Use intelligence to support security investigations and incident response
Data / AI Analysts	Develop and tune stylometric, behavioural, and relationship analysis models
System Administrator	Manage servers, access control, security, backups, and system configuration
Development Team	Design, develop, test, and maintain the software
Testing Team	Test system functionality, performance, security, and reliability

2. EPICS AND USER STORIES

The major requirements of the threat actor de-anonymization system are divided into different Epics. Each Epic represents a large feature area of the system.

Epic 1 – Intelligence Collection and Source Management

User Story 1.1

As an investigator, I want to add authorized dark-web sources, so that the system can collect intelligence from them.

User Story 1.2

As an investigator, I want the system to collect handles, PGP keys, wallet addresses, and timestamps from the configured sources, so that I have raw intelligence for analysis.

User Story 1.3

As an administrator, I want to schedule continuous collection, so that the intelligence stays up to date without manual effort.

Epic 2 – Threat Actor Profile Management

User Story 2.1

As an analyst, I want to create and view threat-actor profiles, so that all known information about an actor is kept in one place.

User Story 2.2

As an analyst, I want to store handles, PGP keys, wallet addresses, and infrastructure indicators against a profile, so that identifiers can be compared across sources.

User Story 2.3

As an analyst, I want to record the source and reliability of every piece of information, so that I can judge how trustworthy it is.

Epic 3 – Relationship and Attribution Analysis

User Story 3.1

As an investigator, I want the system to link personas that share PGP keys, wallet addresses, or infrastructure indicators, so that I can identify related personas.

User Story 3.2

As an analyst, I want AI-based stylometric and behavioural analysis, so that possible links between personas with different identifiers can be detected.

User Story 3.3

As an investigator, I want an attribution confidence score for each relationship, so that I can judge the strength of the evidence.

User Story 3.4

As an investigator, I want to view the evidence behind each link, so that I do not treat a similarity as proof of identity.

Epic 4 – Search, Timeline and Visualization

User Story 4.1

As an investigator, I want to search actors by handle, wallet address, PGP key, or keyword, so that I can find relevant profiles quickly.

User Story 4.2

As an investigator, I want to view activity on a timeline for a selected date range, so that I can follow how personas change over time.

User Story 4.3

As an investigator, I want to view relationships as a graph, so that I can understand connections between actors visually.

Epic 5 – Security, Administration and Reporting

User Story 5.1

As an administrator, I want role-based access and audit logs, so that only authorized users can use sensitive intelligence and every action can be traced.

User Story 5.2

As an investigator, I want to export results in CSV, JSON, and report formats, so that findings can be shared with authorized teams.

User Story 5.3

As an administrator, I want backup and monitoring of the platform, so that data is protected and the system remains available.

3. ACCEPTANCE CRITERIA

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete. The Given–When–Then format is used below.

User Story 1.1 – Add Authorized Source

User Story:

As an investigator, I want to add authorized dark-web sources, so that the system can collect intelligence from them.

Acceptance Criteria:

- Given the investigator is on the source configuration page,
- When the investigator enters valid source details and saves them,
- Then the system should register the source successfully.

- Given the investigator enters an already registered source,
- When the form is submitted,
- Then the system should display an appropriate error message.

User Story 1.2 – Collect Identifiers

User Story:

As an investigator, I want the system to collect handles, PGP keys, wallet addresses, and timestamps from the configured sources, so that I have raw intelligence for analysis.

Acceptance Criteria:

- Given a source is configured and reachable,
- When the collection process runs,
- Then the system should extract and store handles, PGP keys, wallet addresses, and timestamps.

- Given the same identifier has already been collected from the same source,
- When the collection process runs again,
- Then the system should not create a duplicate record.

- Given a source is unreachable,
- When the collection process runs,
- Then the system should log the failure and continue with the other sources.

User Story 2.2 – Store Identifiers on a Profile

User Story:

As an analyst, I want to store handles, PGP keys, wallet addresses, and infrastructure indicators against a profile, so that identifiers can be compared across sources.

Acceptance Criteria:

- Given the analyst opens a threat-actor profile,
- When the analyst adds a valid identifier,
- Then the system should save the identifier with its source and time.

- Given an identifier is already stored for the profile,
- When the analyst tries to add it again,
- Then the system should prevent the duplicate entry.

User Story 3.1 – Link Personas

User Story:

As an investigator, I want the system to link personas that share PGP keys, wallet addresses, or infrastructure indicators, so that I can identify related personas.

Acceptance Criteria:

- Given two personas share the same PGP key or wallet address,
- When the relationship analysis is run,
- Then the system should create a link between the two personas.

- Given no common identifiers exist,
- When the relationship analysis is run,
- Then the system should not create a link.

User Story 3.3 – Attribution Confidence

User Story:

As an investigator, I want an attribution confidence score for each relationship, so that I can judge the strength of the evidence.

Acceptance Criteria:

- Given a relationship has been identified,
- When the investigator opens the relationship,
- Then the system should display a confidence score and the indicators used.

- Given the supporting evidence is weak,
- When the confidence score is calculated,
- Then the system should assign a low confidence level.

User Story 3.4 – View Evidence

User Story:

As an investigator, I want to view the evidence behind each link, so that I do not treat a similarity as proof of identity.

Acceptance Criteria:

- Given a link between two personas exists,
- When the investigator selects the link,
- Then the system should display the matching identifiers, sources, and timestamps.

- Given the link is based only on writing-style similarity,
- When the investigator views the link,
- Then the system should mark it as a possible link that requires verification.

User Story 4.1 – Search Actors

User Story:

As an investigator, I want to search actors by handle, wallet address, PGP key, or keyword, so that I can find relevant profiles quickly.

Acceptance Criteria:

- Given the investigator is logged into the system,
- When the investigator enters a search term,
- Then the system should display the matching threat-actor profiles.

- Given no profile matches the search term,
- When the search is completed,
- Then the system should display a suitable message indicating that no results were found.

User Story 5.1 – Role-Based Access and Audit Logging

User Story:

As an administrator, I want role-based access and audit logs, so that only authorized users can use sensitive intelligence and every action can be traced.

Acceptance Criteria:

- Given a user does not have permission for a feature,
- When the user tries to access it,
- Then the system should deny access and display an appropriate message.

- Given an authorized user views, searches, or exports intelligence,
- When the action is completed,
- Then the system should record the user, action, and time in the audit log.

4. PRODUCT BACKLOG PRIORITIZATION

The product backlog contains all the required user stories. The stories are prioritized based on business value, investigator needs, security requirements, dependencies, and implementation importance.

Priority Levels

- High (P1): Required for the basic working and secure system.
- Medium (P2): Important features that improve the analysis capability of the system.

- Low (P3): Additional features that can be implemented later.

ID	Epic	User Story	Priority
US1.1	Intelligence Collection	Add authorized sources	P1 – High
US1.2	Intelligence Collection	Collect threat-actor identifiers	P1 – High
US2.1	Profile Management	Create and view threat-actor profile	P1 – High
US2.2	Profile Management	Store identifiers on a profile	P1 – High
US5.1	Security & Administration	Role-based access and audit logging	P1 – High
US1.3	Intelligence Collection	Schedule continuous collection	P2 – Medium
US2.3	Profile Management	Record source and reliability	P2 – Medium
US3.1	Relationship Analysis	Link personas using shared identifiers	P2 – Medium
US3.2	Relationship Analysis	AI stylometric and behavioural analysis	P2 – Medium
US3.3	Relationship Analysis	Attribution confidence score	P2 – Medium
US3.4	Relationship Analysis	View evidence behind each link	P2 – Medium
US4.1	Search & Visualization	Search actors by identifier	P2 – Medium
US5.2	Security & Administration	Export CSV, JSON, and reports	P2 – Medium
US4.2	Search & Visualization	Investigation timeline	P3 – Low
US4.3	Search & Visualization	Relationship graph view	P3 – Low
US5.3	Security & Administration	Backup and monitoring	P3 – Low

Reason for Prioritization

Secure access control, source configuration, identifier collection, and profile management are given the highest priority because the system cannot analyze anything until data is collected and stored, and the stored intelligence is sensitive and must be protected from the beginning.

Relationship analysis, AI-based analysis, confidence scoring, and search depend on the collected data, so they are included in the second priority group. Timeline, graph visualization, and backup and monitoring can be developed after the core collection and analysis features are working.

5. SPRINT BACKLOG – SPRINT 1

The first sprint will focus on developing the basic functionality required to protect the system, configure authorized sources, collect threat-actor identifiers, and store them in threat-actor profiles.

Sprint 1 Selected User Stories

User Story	Description	Priority
US1.1	Add authorized sources	High

User Story	Description	Priority
US1.2	Collect threat-actor identifiers	High
US2.1	Create and view threat-actor profile	High
US2.2	Store identifiers on a profile	High
US5.1	Role-based access and audit logging	High

Sprint 1 Task Breakdown

US1.1 – Add Authorized Sources

Tasks:

1. Design source configuration form.
2. Create source database table.
3. Implement source registration logic.
4. Validate source address and access details.
5. Check duplicate sources.
6. Test source configuration.

US1.2 – Collect Threat-Actor Identifiers

Tasks:

1. Design data collection module.
2. Connect the module to configured sources.
3. Fetch and parse posts and profile pages.
4. Extract handles, PGP keys, wallet addresses, and timestamps.
5. Remove duplicate records.
6. Store extracted data with source and time.
7. Handle connection failures.
8. Test collection and extraction.

US2.1 – Create and View Threat-Actor Profile

Tasks:

1. Create threat-actor profile table.
2. Design profile page.
3. Implement create and update logic.
4. Validate profile details.
5. Display profile details.
6. Test profile management.

US2.2 – Store Identifiers on a Profile

Tasks:

1. Create identifier tables for handles, PGP keys, wallets, and indicators.
2. Link identifiers to threat-actor profiles.
3. Prevent duplicate identifiers.
4. Display identifiers on the profile page.
5. Store source and timestamp with each identifier.
6. Test identifier storage.

US5.1 – Role-Based Access and Audit Logging

Tasks:

1. Define user roles.
2. Implement authentication.
3. Implement role-based permissions.
4. Encrypt stored sensitive data.
5. Record user actions in the audit log.
6. Test access control and audit logging.

6. EFFORT ESTIMATION USING STORY POINTS

Story points are used to estimate the relative effort, complexity, and uncertainty involved in completing each user story. The Fibonacci sequence is used for estimation.

The values used are:

1, 2, 3, 5, 8, 13

Higher points indicate greater complexity or effort.

Sprint 1 Story Point Estimation

User Story	Story	Story Points	Justification
US1.1	Add authorized sources	3	Basic form, validation, and database storage are required
US1.2	Collect threat-actor identifiers	8	Most complex because it involves source connection, parsing, identifier extraction, duplicate removal, and error handling
US2.1	Create and view profile	3	Requires a profile table and basic create and view functions
US2.2	Store identifiers on a profile	5	Requires multiple identifier types, linking to profiles, and duplicate checking

User Story	Story	Story Points	Justification
US5.1	Role-based access and audit logging	5	Requires authentication, permissions, encryption, and logging

Total Story Points

3 + 8 + 3 + 5 + 5 = 24 Story Points

Justification of Estimation

Source configuration and profile management are relatively simple features, so they are estimated at 3 points each. Storing different identifier types and implementing access control with audit logging require more design and testing, so they are estimated at 5 points each.

Identifier collection is estimated at 8 points because it depends on source access, requires parsing of unstructured content, extraction of several identifier types, and prevention of duplicate records, and it must handle unreliable sources.

7. SPRINT GOAL AND EXPECTED DELIVERABLES

7.1 Sprint Goal

"To build the foundation of the threat intelligence platform by implementing secure access control, authorized source configuration, threat-actor identifier collection, and threat-actor profile management."

The main objective of Sprint 1 is to create a basic working version of the platform. At the end of the sprint, an authorized user should be able to log in, add a source, collect identifiers from it, and view a threat-actor profile with its stored identifiers.

7.2 Expected Sprint Deliverables

At the end of Sprint 1, the following features are expected to be completed:

- Role-based access control module.
- Audit logging of user actions.
- Authorized source configuration.
- Intelligence collection module.
- Extraction of handles, PGP keys, wallet addresses, and timestamps.
- Threat-actor profile management.
- Identifier storage and display.
- Basic database for sources, profiles, identifiers, and users.
- Duplicate record prevention.
- Basic testing of all Sprint 1 features.

7.3 Definition of Done

A user story will be considered complete when:

- The required functionality is implemented.
- The feature works according to the user story.
- Required data is stored correctly.
- Basic validation is completed.
- Access control and audit logging are applied to the feature.
- The feature is tested successfully.
- Major errors are fixed.
- The feature is ready to demonstrate to stakeholders.

8. SPRINT PLANNING SUMMARY

The Agile approach allows the threat intelligence platform to be developed in small and manageable parts. Instead of developing the entire system at once, the project is divided into multiple sprints.

Proposed Sprint Plan

Sprint	Main Focus	Expected Features
Sprint 1	Foundation, Collection & Profiles	Access control, audit logging, source configuration, identifier collection, threat-actor profiles
Sprint 2	Relationship Analysis & Search	Source reliability, persona linking, evidence view, actor search
Sprint 3	AI Analysis & Automation	Stylometric and behavioural analysis, confidence scoring, scheduled collection
Sprint 4	Visualization, Reporting & Improvements	Timeline, relationship graph, export, backup and monitoring, performance and security improvements

This sprint-based approach allows the development team to receive feedback after every sprint. Problems can be identified early and improvements can be added in the next sprint.

9. EXPECTED BENEFITS OF AGILE SPRINT PLANNING

The Agile sprint planning approach provides several benefits for the threat actor de-anonymization system:

- The system can be developed step by step.
- High-priority security and data-collection requirements are implemented first.
- Investigators and stakeholders can provide feedback after each sprint.
- Errors can be identified and corrected early.

- Changes in requirements, such as new sources or new analysis techniques, can be handled more easily.
- The development team can clearly divide the work.
- Story points help estimate the effort required.
- The product backlog provides a clear list of pending requirements.
- Sprint goals help the team remain focused.
- Working software can be delivered at the end of every sprint.

10. CONCLUSION

The Agile Sprint Planning and User Story Workshop provides a structured way to develop the AI-assisted Dark Web Threat Actor Intelligence & Attribution Platform. The project vision identifies the main purpose and stakeholders, while Epics and User Stories divide the requirements into smaller and understandable features.

Acceptance criteria help the development and testing teams understand when a feature is complete. The prioritized product backlog identifies the most important requirements, and the Sprint 1 backlog focuses on secure access, source configuration, identifier collection, and profile management. Story point estimation helps the team understand the effort required for each user story.

The selected Sprint 1 goal is to build the foundation of the platform. Further sprints can add relationship analysis, AI-based analysis, confidence scoring, timeline and graph visualization, reporting, and other improvements. This approach makes the project easier to develop, test, maintain, and improve based on stakeholder feedback.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25